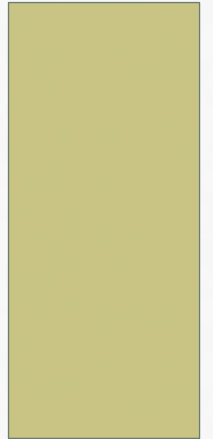


CLASS IV TOOTH PREPARATION AND RESTORATION



Learning outcomes

- Explain in detail about the steps in class IV cavity preparation.
- Explain in detail about the steps in restoration of class III and IV cavity preparation.

CLASS IV RESTORATIONS

- Restorations on the proximal surfaces of anterior teeth that do involve the incisal angle



Class IV Tooth Preparation

- The Class IV composite restoration has provided the dentist with a conservative treatment to restore fractured, defective, or carious anterior teeth
- Preoperative assessment of occlusion is very important





- Shade selection is the first step in composite restoration



- Preparation is similar to Class III except that the preparation for class IV is extended to the incisal angles

Tooth preparation

- Conventional
- Beveled conventional
- Modified

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Conventional class IV

- The conventional tooth preparation design similar to class conventional III has minimal clinical Class IV application
- Used in large class IV caries which extending in the root.

BEVELED CONVENTIONAL CLASS IV

- Indicated for restoring large proximal areas that involve incisal angle.



The cavosurface bevel provide additional retention
It is best prepared with a flame shaped diamond bur, in an angle approximately 45 degrees to the external tooth surface

The width of the bevel should be 0.5 to 2 mm, depending on the amount of tooth structure missing and the retention perceived necessary.

- Primary retention is obtained by Micromechanical bonding
- Additional(Secondary) retention can be provided by

Bevels



Dovetail extensions

A dovetail extension onto the lingual surface of the tooth may enhance both the restorations strength and retention, but it is less conservative and therefore not used often.



MODIFIED CLASS IV

- Indicated for small to moderate class IV lesions
- Main objective to remove little tooth structure as possible
- Used commonly in minor traumatic injuries, Fracture line limited to enamel
- Little tooth Preparation is required
- Only Cavo-surface margin is beveled.



RESTORATIVE TECHNIQUE FOR CLASS III AND IV

1. Matrix band and wedge application

A properly contoured matrix and wedge application is a prerequisite for a restoration involving the entire proximal contact area.

A proper matrix adaptation reduces the amount of excess material, minimizing the finishing time.

A matrix for the proximal surface of an anterior tooth should be made of a thin material, such as polyester, that can be contoured easily.

MATRIX BAND FOR CLASS III & IV

- A Clear polyester strip matrix or mylar strip or celluloid strip
- A properly contoured polyester strip matrix is used for most Class III and IV preparation
- The matrix is extended at least 1 mm beyond the prepared gingival and incisal margins.



- Sometimes the matrix strip does not slide through because of tight contact . In such instances, a wedge is lightly positioned in the gingival embrasure before the strip is inserted.



WEDGE

- A wedge is needed at the gingival embrassure to
 - (1) To hold the matrix strip in position
 - (2) Provide slight separation of the teeth
 - (3) To prevent a gingival overhanging of the composite material.

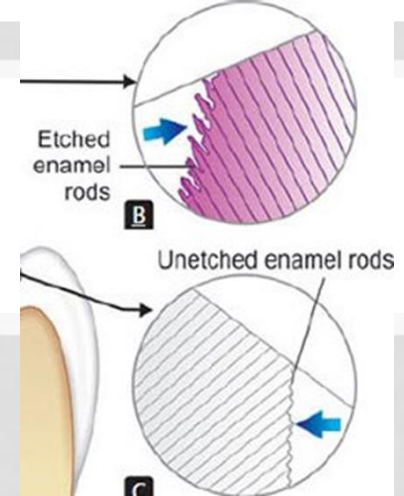


- The wedge is placed, using tweezers from the facial approach for lingual access preparations and from lingual side for facial access preparation, just apical to the gingival margin.



2. Acid etching –

- Increases surface area
- Etching helps in micromechanical bonding – more resin penetrate into micro porosities.
- DEVELOPED BY BUNOCORE IN 1955
- Agent used - 37 % PHOSPHORIC ACID



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- Apply the gel etchant 0.5 beyond the prepared margins .

1. Total acid etching – Complete enamel and dentin etching – 15 seconds

2. Selective acid etching- First enamel is etched for followed by dentin etching

- Enamel etching – 30 seconds
- Dentin etching – 15 seconds



- The area is washed with water to remove the etchant. Same amount of time as etching time.

- Dry the tooth structure, if dentin is exposed, do not air dry.
- Use small cotton pellet, disposable brush or tissue paper to remove excess water.
- White frosted appearance should appear after etch and dried.



3. Adhesive application

Bonding system is applied on all tooth structure that has been etched with a microbrush

Light Cure as described by manufacture in a range of 20 seconds



4. Insertion and curing of composite

- Insertion of composite in incremental manner not more than 2 mm to minimize polymerization shrinkage
- When the material has been inserted, it is cured through the strip for 20 seconds
- One should not touch the strip with the tip of the light initially because it could distort the contour of the restoration.

5. Contouring and finishing of composite

Good technique and experience in inserting the composites significantly reduce the amount of finishing required.

Coarse diamond burs can be used to remove gross excess, but they generally are not recommended for finishing composites, because of the high risk of inadvertently damaging contiguous tooth structure.



CONTOURING THE CONVEX AREA

- Soflex disks (3M) or Supersnap disks (Shofu) is available for contouring as well as polishing the convex areas such as labial areas and flat surfaces.
- Oval shaped Finishing burs are more suitable for concave surfaces
- These disks are flexible and are produced in several diameters and abrasive textures.



- Regardless of the type of disk chosen, disks are used sequentially from coarse to fine, generating a smooth surface.
- The external enamel surface should act as a guide for proper contour and surface texture.



- Interproximal and Embrasure Areas.

The proximal surface contours and margins should be assessed visually and tactilely with an explorer and dental floss.

By passing the floss in between the contact, one should hear snap sound and “shoe-shined” as it is pulled occlusally.

If the floss catches or frays, additional finishing is indicated.

HOW TO REMOVE PROXIMAL OVERHANGS

A No.12 surgical blade in a Bard-Parker handle may be used because the curved shape of the blade and the thin diameter make this instrument ideal for removing gingival overhangs.



INTERPROXIMAL FINISHING

- Overhangs at the gingival(proximal)area can also be removed by Narrow finishing strips may be used to smooth the gingival proximal surface



- Some strips have two different types of abrasives (medium and fine) on opposing ends of the strip, with a small area between where no abrasive is present to allow easy and safe insertion of the strip through the contact area.

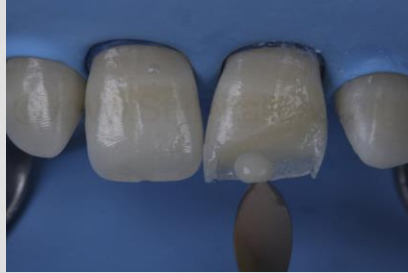
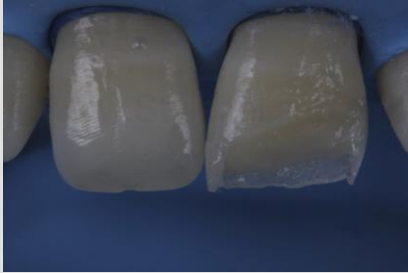
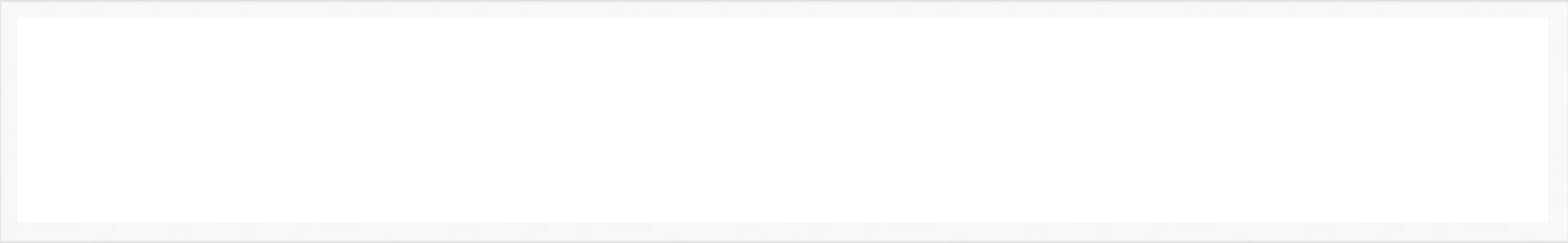


6. Polishing of the composite

- The restorations are finished with appropriate polishing points, cups, spirals .







- For most Class IV preparations, a polyester strip matrix or other matrix is used as previously described in this presentation.
- For large Class IV tooth preparations, A custom lingual matrix made by using putty-like polyvinylsiloxane impression material or a quick-set silicone matrix material, such as Template .





Essential reading:

- Sturdevant's 5th edition art and science of operative dentistry page no- 541 -551
- Pickard's Manual of Operative Dentistry, 8 th edition page no- 156-159